ICE COTTON NO. 2 FUTURES (CT)

Spread & Curve Structure — Fundamental Reference

Crop Production, China, Synthetic Substitution, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of ICE Cotton No. 2 futures (ticker: CT), the global benchmark for upland cotton traded on the Intercontinental Exchange (ICE Futures US, formerly NYBOT). Cotton is unique among commodity futures in that it competes directly with synthetic fibres — primarily polyester — for the same end-use market (textiles). This competition means that cotton demand is not simply a function of income and population growth but also of the relative price of petroleum-derived fibres, which in turn links cotton to crude oil markets. At the same time, cotton supply is determined by planted acreage decisions that compete with corn, soybeans, wheat, and other crops for the same farmland — embedding cotton in the broader row-crop complex. China, as the world's largest cotton consumer and a major but declining producer, plays a uniquely pivotal role: its state reserve management programme, import quota decisions, and domestic price support policies collectively make Chinese policy the most powerful single non-weather variable in the global cotton balance. Understanding all these dimensions is the foundation of spread and butterfly trading on the CT curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

ICE Cotton No. 2 is a physically delivered contract. Delivery is made via shipping certificates issued by certified warehouses in specific US producing regions, typically in the Southeast and Mid-South. The delivery grade structure and the certified warehouse system create a specific basis relationship between physical cotton markets and the futures price.

Parameter	Detail
Exchange	ICE Futures US (formerly NYBOT)
Ticker	CT (front month)
Full name	Cotton No. 2
Contract size	50,000 pounds net weight (approximately 100 bales of 500 lb each)
Quotation	US cents per pound (¢/lb); minimum tick = 0.01¢/lb = $5.00 per contract
Contract months	March (H), May (K), July (N), October (V), December (Z) — five delivery months per year. Liquid trading concentrated in front 3–4 contracts; December is the most liquid (new-crop harvest month).
Settlement type	Physical delivery via electronic shipping certificate issued by ICE-certified warehouses in Memphis, TN; Greenville, SC; Galveston, TX; and other approved locations.
Delivery grade	Middling 1-3/32" (the US grade standard). Above Middling and below Middling grades are deliverable at schedule premiums and discounts. Colour, leaf, and preparation standards also apply (USDA HVI — High Volume Instrument classification).
Last trading day	17 business days from the end of the delivery month (approximately the third week of the delivery month's final session)
Trading hours	ICE electronic: Sunday–Friday 21:00–14:00 ET; most liquid approximately 09:00–14:00 ET
Contract value at 80¢/lb	$40,000 per contract
Price limits	$0.04/lb ($2,000) initial daily limit; expandable
Marketing year	August 1 – July 31 (US cotton marketing year). The December contract is the primary new-crop contract; July is the last old-crop contract.

1.1 The Old-Crop / New-Crop Structure

The distinction between old-crop and new-crop contracts is the most important structural feature of the CT curve. The US cotton marketing year runs August 1 to July 31. Within any given year:

- **Old-crop contracts (March, May, July)**: price the cotton from the previous harvest that is currently in the warehouse system and being traded. Old-crop supply is fixed — the crop has been harvested. Old-crop pricing is therefore determined primarily by existing physical stocks, current demand pace, and export shipment rates.
- **New-crop contracts (December of current year, March of following year)**: price the cotton from the harvest that has not yet occurred (or is underway). New-crop supply is uncertain and determined by planting intentions, weather, and yield. The December contract is the primary new-crop reference.
- **The old-crop/new-crop spread (July-December)**: the spread between the last old-crop contract (July) and the first new-crop contract (December) is one of the most watched structural spreads in cotton. When July trades at a large premium to December, the market is in old-crop tightness (existing stocks are short relative to demand). When December trades at a premium to July (inverted

old-crop/new-crop), the market anticipates new-crop relief after a tight old-crop period.

1.2 The HVI Grading System and Delivery Quality

- **High Volume Instrument (HVI) classification**: USDA grades US cotton using automated HVI instruments that measure fibre length, strength, micronaire (fineness/maturity), length uniformity, colour (reflectance and yellowness), and leaf content. These quality parameters determine the grade premium or discount relative to the Middling 1-3/32" base grade.
- **Micronaire (mic)**: a combined measure of fibre fineness and maturity. The "ideal" mic range for most mill applications is 3.7–4.2. Cotton below 3.5 (immature) and above 4.9 (coarse) faces processing difficulties and price discounts. Micronaire is affected by weather during the boll development period (cool late season = low mic; hot and water-stressed = high mic). Anomalous mic distributions in the US crop can affect the quality of deliverable supply.
- **Staple length**: fibre length is a key quality parameter. Extra-long staple (ELS) cotton — Egyptian Giza, US Pima (Gossypium barbadense species) — is a premium product used for fine fabrics and does not deliver against the No. 2 contract (which is standard upland, Gossypium hirsutum). Upland staple lengths of 1-3/32" (Middling) to 1-5/32" (Strict Middling) are the primary delivery range.

2. Global Production — The Supply Landscape

World cotton production is approximately 25–27 million metric tonnes (110–120 million 480-lb bales) per year. Supply is less geographically concentrated than sugar or cocoa — cotton is grown across approximately 80 countries — but the dominant producers are well-defined.

Country	Production (approx.)	Harvest Period	Key Supply Risk	CT Curve Impact
India	5.5–7.0 Mb	Nov–Mar ginning; Kharif crop (Jun–Oct growing)	Monsoon (ENSO); pink bollworm; whitefly; minimum support price (MSP) affecting area	Largest producer. Indian production revisions drive multi-month CT repricings. MSP relative to CT determines whether Indian farmers maintain or reduce cotton area.
China	5.5–6.5 Mb	Sep–Nov harvest; Xinjiang dominant (85%+ of crop)	Water scarcity in Xinjiang; forced labour scrutiny (supply chain bans); government reserve policy	Declining producer but largest importer/consumer. Chinese reserve policy (buy/sell decisions) creates sudden supply/demand shifts. Xinjiang supply chain bans create quality/origin differentiation.
United States	13–18 Mb	Aug–Nov harvest; ginning Oct–Feb	Drought (Texas High Plains — most critical); hurricane (SE); heat stress; boll weevil (eradicated but monitoring continues)	Price-setter for CT futures; US crop is the primary deliverable supply. Texas accounts for ~45% of US crop; West Texas drought is the dominant near-term supply variable.
Brazil	13–16 Mb	Mar–Jun harvest in Mato Grosso (tropical savanna)	Drought; soil moisture at planting; logistics (Mato Grosso landlocked — long truck distances to Santos port)	Fast-growing exporter. Brazil has been the world's second-largest exporter. Mato Grosso production revisions affect the global export balance and CT mid-curve.
Pakistan	5.5–7.5 Mb	Sep–Dec ginning; Jul–Oct growing	Monsoon flooding (devastating in 2022); cotton leaf curl virus; water availability from Indus system	Significant producer but domestic mill demand typically exceeds production — Pakistan is usually a net importer. Production shortfalls directly affect domestic spinning availability.
Australia	4.0–6.0 Mb	Mar–May harvest; Nov–Feb growing	La Nina: good rainfall for irrigation recharge (bullish for area); El Nino: drought reduces irrigation water	High-quality Australian cotton (generally Strict Middling to Good Middling grade). Counter-seasonal supply to US. Growing exporter to Asian mills.
Turkey, Greece, Uzbekistan, others	2–5 Mb combined	Sep–Nov harvest	Water availability; government policies; Uzbekistan forced labour history	Regional supply for European and Central Asian mills. Uzbekistan cotton historically excluded from some markets due to forced labour, though reforms have been implemented.

The cotton marketing year runs August 1 to July 31 (USDA convention). The global balance sheet is compiled on this basis by USDA WASDE and ICAC. Production from different hemispheres arrives at different times: US and Northern Hemisphere production enters the market October-November; Brazilian and Australian production March-June. This staggered production calendar shapes the seasonal spread structure.

3. The United States Crop — The Primary Deliverable Supply

The US cotton crop is the direct supply source for the ICE Cotton No. 2 futures contract — it is the primary deliverable grade and origin. Understanding the US crop calendar, regional production structure, and key weather risks is essential for interpreting CT spread movements.

3.1 US Cotton Belt Regions

Region	States	Share of US Crop	Dominant Risk	Typical Grade	CT Impact
Texas High Plains (Rolling Plains)	West Texas (Lubbock, Plainview)	~40–45%	Dryland production — entirely rain-dependent. Severe drought years (2011, 2022) can reduce TX HP production by 50–70% from a good year. Hail. Early frost.	Middling to Strict Middling. Lower mic common in dry years.	Single most important regional supply variable. TX HP abandonment rate (% of planted acres not harvested) is the most watched crop-condition metric each August-September.
Southeast (Carolinas, Georgia)	NC, SC, GA, AL, FL	~15–20%	Hurricane (track and timing relative to boll opening is critical). Excess moisture during harvest. Nematodes.	Generally good quality; Middling to Strict Middling.	Hurricane risk in August-September can destroy standing bolls before harvest. Quality impacts (weather-damaged grades) after storm events.
Mid-South (Delta)	MS, AR, TN, MO, LA	~15–20%	Excessive rainfall at planting and during boll development. Late-season rain damage at harvest.	Good quality; well-suited to Middling grade. Proximity to Memphis warehouse delivery.	Important for deliverable supply into Memphis warehouses — the primary CT delivery point.
Southwest (Oklahoma, south TX)	OK, southern TX	~8–12%	Drought; heat stress. Partly irrigated.	Variable — drier conditions often produce higher mic.	Moderate. Contributes to overall US balance.
Far West (California, Arizona, NM)	CA, AZ, NM	~3–5%	Water availability (Colorado River allocation critical). High production cost.	High quality; longer staple. Pima (ELS) dominant in CA.	Pima does not deliver against CT. Upland CA/AZ is small volume. California water crisis is a structural long-run supply reduction signal.

3.2 The US Cotton Crop Calendar

Period	Stage	Key Weather Risk	Market Data Release	CT Impact
Feb–Mar	Planting intentions	Winter soil moisture conditions	USDA Prospective Plantings (late March)	First official acreage signal for new-crop December CT

Apr–May	Planting	Cool, wet soils delay planting; late planting = late maturity = frost risk	USDA weekly Crop Progress (planted %)	Planting pace affects crop maturity timeline and frost risk
May–Jun	Emergence; squaring	Adequate moisture critical; flooding in Mid-South	USDA Crop Progress (emerged, condition)	Early season condition ratings establish yield potential baseline
Jul–Aug	Boll development; bloom	Heat stress (above 95F reduces boll set); drought; hail; storm tracks	USDA Crop Progress weekly; NASS June Acreage Report	Most critical weather window. August condition ratings most market-moving.
Aug–Sep	Boll opening; harvest beginning in TX	Hurricane (Southeast); early frost (High Plains); excessive rain delays boll opening	USDA August WASDE (first production estimate); NASS August crop report	First WASDE production estimate is the most market-moving scheduled event of the crop year
Oct–Nov	Peak harvest; ginning begins	Frost; excessive moisture degrades picked cotton quality	USDA WASDE monthly updates; NASS ginning reports	Production estimates refined monthly; crop quality data from HVI classification
Dec–Feb	Ginning; marketing; export loading	No growth risk; logistics/infrastructure risk (port congestion, truck/rail)	USDA weekly export inspections; weekly export sales; NASS ginning progress	Export pace vs USDA forecast determines whether the ending stocks estimate is achieved

3.3 The Texas High Plains Drought Risk

The Texas High Plains (THP) is the single most important supply variable in the US cotton market and therefore in CT futures. The region accounts for approximately 40–45% of US production but has extremely variable rainfall — it is at the margin of dryland farming feasibility:

- **Dryland production**: approximately 75–80% of THP cotton is dryland (no irrigation). In drought years, dryland cotton produces very little or is abandoned. The abandonment rate (percentage of planted acres that are not harvested due to poor stands) is the single most watched crop metric in late summer.
- **2011 drought**: the most severe documented Texas drought in modern history. US cotton production fell from approximately 18.3 Mb to 15.6 Mb. Texas abandonment reached 70%+ in some counties. CT rallied to record highs above 200¢/lb in early 2011 (a confluence of global demand and US supply disruption).
- **2022 drought**: another severe THP drought reduced US production significantly. Texas abandonment rates again exceeded 60% in dryland areas. CT peaked above 150¢/lb.
- **Ogallala Aquifer depletion**: the primary irrigation water source for THP cotton is the Ogallala Aquifer, which is being depleted faster than natural recharge. As Ogallala water levels decline, irrigated THP cotton area contracts structurally — a slow-moving long-term bearish supply signal for the CT back curve.
- **Palmer Drought Severity Index (PDSI)**: the key weather metric for THP conditions. Published weekly by NOAA. A PDSI below -3 (severe drought) in June-July is an acute bullish signal for the front CT contract.

4. China — The Pivotal Consumer and Policy Actor

China's role in global cotton markets is unlike that of any other single actor in any major commodity. As both the world's largest cotton consumer (approximately 8–9 million metric tonnes per year, roughly 35% of global mill use) and a major producer, China's domestic policy decisions create demand and supply shocks that propagate directly through the CT futures curve.

4.1 Chinese Cotton Consumption

- **Mill use**: China's textile and apparel industry consumes more cotton than any other country. The cotton spinning industry is centred in Shandong, Jiangsu, Xinjiang, Zhejiang, and Guangdong provinces. Chinese cotton consumption is sensitive to: GDP growth (domestic apparel demand), export orders from Western brands (the US-China trade relationship affects apparel exports), and the relative price of cotton vs polyester (synthetic substitution is more prevalent in Chinese mills than in US or European mills).
- **Cotton-polyester substitution in China**: Chinese textile mills are typically smaller-scale and more price-sensitive than Western mills. When cotton prices are high relative to polyester (which is petroleum-derived and tracks PTA/MEG/crude oil prices), Chinese mills shift fiber mix toward higher polyester content. This substitution can reduce Chinese cotton demand by 500,000–1,000,000 MT in extreme price scenarios.
- **Chinese yarn and fabric exports**: China exports cotton yarn, fabric, and apparel to global markets. When global apparel demand softens (recession, trade barriers), Chinese mill utilisation falls and cotton demand weakens.

4.2 Chinese State Reserve System — The Most Powerful Policy Variable

China operates a state cotton reserve managed by the China National Cotton Reserve Corporation (CNCRC). The reserve was established to support domestic cotton farmers and ensure domestic supply security. Reserve accumulation and release decisions are the single most powerful policy variable in global cotton:

- **Reserve accumulation (2011–2013)**: China aggressively purchased domestic and imported cotton to build the state reserve, at times holding an estimated 50–60% of global annual production in storage. This programme was simultaneously bullish for CT (the buying drove prices higher) and ultimately bearish (when the reserve was eventually sold).
- **Reserve sales (2014–2018)**: China began selling cotton from the state reserve via weekly auctions to domestic mills. The overhang of reserve supply suppressed Chinese import demand for years and weighed on the CT mid-to-back curve. The reserve was partially destocked but the precise remaining reserve size is not publicly disclosed.
- **Import quota system**: China allocates import quotas for cotton: a 1% tariff-rate quota (TRQ) of 894,000 MT/year, and above-quota imports at 40% tariff. The 40% tariff effectively sets a price ceiling at which Chinese mills switch to domestic cotton, reserve cotton, or synthetic fibres rather than importing above-quota. Annual decisions to issue additional above-quota licences (sliding-scale tariff certificates) are market-moving events.
- **Chinese cotton import volumes**: typically 1.5–3.5 Mb/year depending on domestic production, reserve levels, and price relationships. Major origins: US, Australia, Brazil, Burkina Faso, Mali. The US-China trade relationship (Section 301 tariffs) has periodically disrupted US cotton exports to China, redirecting Chinese import demand to Brazil and Australia.

4.3 Xinjiang Cotton and Supply Chain Complexity

- **Xinjiang produces approximately 85% of Chinese cotton** and approximately 20–25% of global production. Since 2020, Xinjiang cotton has faced significant scrutiny due to forced labour allegations in the region (related to the treatment of Uyghur minorities).

- **US Uyghur Forced Labor Prevention Act (UFLPA, effective June 2022)**: creates a rebuttable presumption that goods from Xinjiang (or made with Xinjiang inputs) are produced with forced labour and are therefore barred from US import. Apparel brands and retailers have responded by shifting sourcing away from Xinjiang cotton, creating a two-tier global cotton market: Xinjiang-origin (primarily accessible to non-Western supply chains) and non-Xinjiang certified (accessible to Western brands).
- **Traceability and certification**: systems like the Textile Exchange standards, Cotton LEADS, and GOTS certification are growing rapidly as brands seek to document the origin of cotton used in their products. Certified non-Xinjiang cotton (US, Australia, Brazil, India with certification) commands a premium in certain supply chains.
- **Market impact**: the Xinjiang issue primarily affects quality/origin differentials and trade flow patterns rather than the absolute CT futures level — but it has increased demand for US, Australian, and Brazilian cotton specifically (all are Western-supply-chain compatible), which is structurally supportive for CT prices relative to a world where Xinjiang cotton was freely traded everywhere.

5. India — The Largest Producer and a Key Exporter

India is the world's largest cotton producer by area (approximately 12–13 million hectares planted) and competes with China for the title of largest total producer. Indian production and export policy are the second most important supply variable after the US crop.

5.1 Indian Cotton Production

- **Kharif crop**: cotton is a Kharif (summer-monsoon) crop in India, planted June–July with the monsoon onset and harvested October–March (ginning peak November–February). The Southwest Monsoon is therefore the primary weather variable — adequate monsoon rainfall determines both planted area (late monsoon reduces area) and yield.
- **Pink bollworm resistance**: Bt (Bacillus thuringiensis) cotton was introduced in India in 2002 and dramatically increased yields by controlling bollworm. However, pink bollworm developed resistance to Bt protein (documented from 2014 onward), reducing the efficacy of Bt cotton and increasing yield losses. The spread of bollworm resistance is a slow-moving structural bearish signal for Indian yields unless new technology is introduced.
- **Whitefly and other sucking pests**: whitefly infestations have damaged Indian cotton crops (particularly in Punjab) multiple times since 2015. Pesticide resistance in whitefly populations has made control more difficult.
- **Minimum Support Price (MSP)**: the Indian government sets an MSP for cotton at the gin (Minimum Support Price for Fair Average Quality). When the MSP is above the market price, Cotton Corporation of India (CCI) is mandated to purchase cotton. A high MSP relative to CT incentivises Indian farmers to maintain cotton area even when world prices are low, creating a supply floor for Indian production.

5.2 Indian Cotton Exports

- **Export volumes**: India typically exports 4–7 Mb/year, primarily to Bangladesh, Vietnam, Indonesia, and China. Indian exports compete directly with US exports in Asian markets.
- **Export restrictions**: India has periodically imposed export restrictions or minimum export prices (MEP) to protect domestic mill interests (keeping domestic cotton available and affordable for Indian spinning mills). Announcements of Indian export restrictions are immediately bullish for CT (removes supply from global market) while removal of restrictions is bearish.
- **India-Pakistan trade**: the India-Pakistan relationship affects cotton trade in the region (Pakistan is a natural market for Indian cotton, being geographically adjacent and having a large spinning industry). Diplomatic tensions periodically disrupt this trade, forcing Pakistan to source cotton from more distant origins.

6. Cotton vs Polyester — The Synthetic Fibre Competition

Cotton's most distinctive fundamental feature is its direct competition with polyester and other synthetic fibres for the textile market. This competition makes cotton demand a function not only of income and population growth but also of petroleum prices, since polyester is manufactured from purified terephthalic acid (PTA) derived from para-xylene (a petroleum-derived aromatic).

6.1 The Global Fibre Market

- **Total global fibre consumption**: approximately 110–115 million tonnes/year. Cotton's share has declined from approximately 50% in 1990 to approximately 22–25% today, with polyester now accounting for approximately 55–60% of total fibre. This structural share loss is the most important long-run demand headwind for cotton.
- **Polyester production cost**: polyester (PET fibre) is manufactured from monoethylene glycol (MEG, derived from ethylene) and purified terephthalic acid (PTA, derived from para-xylene). Para-xylene is a petroleum refinery product. Therefore, polyester production cost is ultimately linked to crude oil prices — though the correlation is imperfect and mediated by refinery economics, para-xylene margins, and PTA overcapacity in China.
- **The cotton-polyester price ratio**: when cotton trades at a significant premium to polyester (on a fibre-equivalent basis), textile mills — particularly in China and South/Southeast Asia — substitute toward blended fabrics (cotton/polyester mix) or fully synthetic fabrics. This substitution reduces cotton demand and provides a price ceiling of sorts: cotton cannot rise far above polyester parity without demand destruction.
- **The inverse relationship**: when oil prices are high and polyester is expensive, cotton becomes relatively more price-competitive, supporting cotton demand and CT prices. When oil prices collapse, cheap polyester directly pressures cotton demand and CT. 2014–2016 (oil price collapse) was a period of severe cotton demand pressure from cheap polyester.

6.2 Blending Ratios and Consumer Preferences

- The share of cotton in blended textiles responds to relative prices over a 3–12 month lag, as mills adjust raw material purchasing and blend specifications. A 10% increase in the cotton-polyester price ratio has been estimated to reduce cotton's share of blended fabrics by 1–3 percentage points over a 6-month period — a modest but real demand elasticity.
- Consumer preference for natural fibres (sustainability, comfort) in developed markets provides some structural demand support at the premium end. Fast fashion, which prioritises low cost over fibre type, is predominantly polyester.
- **Recycled polyester** (rPET, from recycled PET bottles) is a growing alternative that is marketed as sustainable and competes with both virgin polyester and cotton in the sportswear and outdoor segment.

7. Demand Fundamentals

Global cotton mill use is approximately 25–27 million metric tonnes per year (approximately 115–125 million 480-lb bales), growing at approximately 1–2% per year on average — well below the growth rate of synthetic fibres.

7.1 Mill Use by Region

Region	Mill Use (approx.)	Cotton Share of Fibre Mix	Primary Driver	CT Curve Sensitivity
China	8–9 Mb/yr	~30–35%	Apparel export orders; domestic demand; state reserve policy; cotton-polyester price ratio	Dominant — any shift in Chinese mill use moves the global balance. Cotton-polyester spread is most relevant for Chinese demand.
India	5–6 Mb/yr	~60%+	Domestic apparel demand; export yarn/fabric orders from Bangladesh/Vietnam; MSP relative to cotton price	High cotton content due to strong Indian preference for natural fibre. Indian mill use is relatively stable.
Pakistan	2.5–3.5 Mb/yr	~70%+	Yarn/fabric export orders (primarily to EU under GSP+ preferences); electricity and gas availability for mills	Pakistan is a large cotton spinner dependent on imports when domestic production falls short. Energy crises affect Pakistani mill utilisation.
Bangladesh	1.5–2.0 Mb/yr	~50–60%	Garment export orders (world's 2nd largest apparel exporter); imported yarn from India/China	Bangladesh imports most of its cotton as yarn or raw. Garment export orders from Western brands are the primary demand signal.
Turkey	1.2–1.5 Mb/yr	~45–55%	EU apparel export orders; domestic textile demand; cotton-polyester price	Moderate. Turkey is a sophisticated textile producer with diversified fibre use.
Vietnam, Indonesia, others (SE Asia)	2.0–3.0 Mb combined	~35–50%	Garment and textile export manufacturing for global brands	Growing rapidly. Supply chain diversification away from China has increased cotton demand in Vietnam and Indonesia.
USA, EU, Brazil	~2.0 Mb combined	~35–50%	Domestic textile/apparel demand; industrial uses (non-wovens, medical)	Declining share of global mill use. US domestic mills use primarily US-grown upland cotton.

7.2 The Apparel and Textile End Market

- **Global apparel retail sales**: the ultimate demand driver for cotton. Consumer spending on clothing is correlated with GDP, consumer confidence, and employment. Recession reduces discretionary apparel spending; recovery restores it. The COVID-19 pandemic caused a sharp 2020 demand contraction followed by a strong 2021 rebound.
- **Brand sourcing decisions**: major apparel brands (Nike, H&M;, Zara parent Inditex, PVH, VF Corporation) drive sourcing decisions that determine which countries' cotton flows into the global supply chain. Brand sustainability commitments (Better Cotton, organic cotton, recycled fibre

targets) affect cotton quality premiums and certain origin preferences.

- **The supply chain lag**: raw cotton → yarn → fabric → garment → retail takes approximately 6–9 months. CT futures pricing therefore incorporates both current physical demand and anticipated demand 6–9 months forward. Retail sales data and brand order books are leading indicators that the physical market translates into cotton demand after this lag.
- **Bangladesh and Vietnam as demand transmission points**: both countries are major garment exporters that import cotton and yarn as raw material. Their export order books (reported via industry associations BGMEA in Bangladesh, VITAS in Vietnam) provide a mid-curve demand signal.

8. The Cotlook A Index and the Physical Cotton Market

The Cotlook A Index is the primary global benchmark for physical cotton prices — analogous to Dated Brent for crude oil. Understanding its relationship to CT futures is essential for interpreting basis and spread behaviour.

8.1 The Cotlook A Index

- **What it measures**: Cotlook Ltd (UK) publishes a daily assessment of the cost-and-freight (CIF) value of cotton at Far East ports (primarily China, but also Bangladesh, Vietnam, India). It is calculated as the average of the five cheapest origins offering cotton CIF Far East on that day. It includes US, Australian, Brazilian, West African, and other origin cotton.
- **Cotlook A vs CT futures**: the Cotlook A Index is CIF (cost, insurance, freight) while CT is a FOB US delivery price. The Cotlook A tends to trade at a premium to CT of approximately $5–15/bale in normal conditions (reflecting freight from US to Far East, plus the value of non-US origins that have different logistics costs). When the Cotlook A is at a large premium to CT, US cotton is cheaply priced relative to global alternatives — export demand for US cotton increases. When CT is close to or above the Cotlook A equivalent, US cotton loses export competitiveness.
- **The A-B spread (Cotlook A minus Cotlook B)**: the B Index measures lower-quality cotton. The A-B spread indicates the premium for quality and is a proxy for the global supply/demand balance at different quality levels.

8.2 US Export System — USDA Weekly Data

- **Weekly Export Sales Report (ESR)**: USDA FAS publishes weekly net export sales and shipments for upland and ELS cotton. The upland cotton export sales data (new sales bookings and actual shipments) is the most important weekly demand data release. Strong weekly sales to China or Bangladesh confirm demand and are bullish for CT. Cancellations (negative net sales) are bearish.
- **Cumulative pace vs USDA forecast**: USDA publishes a full-year export forecast in the WASDE each month. The market tracks whether actual weekly sales and shipments are "on pace" to achieve the forecast. Falling below pace requires a WASDE downward revision (bearish for CT mid-curve); running above pace requires an upward revision (bullish).
- **Export destinations**: the ESR breaks out sales by destination country. China, Bangladesh, Vietnam, and Pakistan are the primary destinations for US upland cotton. A shift in the mix of buyers — e.g. China absent due to tariffs, replaced by more Bangladesh/Vietnam — can affect the pace and profitability of US exports.
- **Certified warehouse stocks**: ICE publishes daily data on cotton certified for delivery against the futures contract. High certified stocks are bearish (supply available for delivery is plentiful); low certified stocks are bullish (delivery squeeze risk if short sellers are present). The relationship between certified stocks and open interest in the front contract determines squeeze risk near expiry.

9. The Forward Curve Structure — Zones and Spread Mechanics

The CT forward curve has five contract months per year (March, May, July, October, December), creating spreads of varying durations. The most important structural divide is old-crop (March-May-July) vs new-crop (October-December), with the July-December spread as the primary old-crop/new-crop transition signal.

9.1 Contract Characteristics and Spread Drivers

Contract	Crop Year Role	Primary Driver	Secondary Driver	Key Market Signal
December (Z)	New-crop anchor. First large contract after the new US harvest enters the market.	US planted acreage; weather during boll development (Jul-Aug); Texas HP drought and abandonment; first WASDE production estimate (August)	Global supply/demand balance at end of prior year; early Brazilian crop outlook	USDA Prospective Plantings (March); NASS June Acreage; weekly Crop Progress Aug-Sep; August WASDE
March (H)	New/old crop bridge. Post-harvest. Active export season.	US crop quality and supply (harvest complete); export sales pace vs USDA forecast; Brazilian harvest early indications (Mar-Jun)	Chinese import demand; certified warehouse stocks level	USDA weekly export sales and shipments; certified stock level; WASDE monthly revision
May (K)	Old-crop. US marketing year approaching end. Brazilian harvest peak.	US ending stocks projection; Brazilian cotton export competition beginning; global balance for the marketing year	Indian export policy; Pakistani import demand	USDA WASDE May-July revisions; ICAC balance sheet; Brazilian CONAB estimates
July (N)	Last old-crop contract. New-crop planting underway in US.	Old-crop/new-crop spread relationship; US ending stocks; early new-crop planting condition	Southern hemisphere (Australian harvest Mar-May); US crop progress pace	Certified stock vs July open interest (delivery squeeze indicator); USDA weekly crop progress; Australian ABARES estimates
October (V)	New-crop. US harvest in early stages. Global balance for new year establishing.	US harvest progress and condition; early production estimates; global new-year demand outlook (China, Bangladesh orders)	Indian kharif crop condition (monsoon outcome); Brazilian new-season planting intentions	USDA September-October WASDE; NASS October production estimate; weekly US harvest progress

9.2 The July-December Spread (Old Crop / New Crop)

The July-December spread is the most structurally informative spread in the cotton market:

- **July at large premium to December (steep backwardation)**: old-crop supply is tight. Physical cotton is scarce; the market values available cotton now over new-crop cotton that has not yet been harvested. This typically occurs when US ending stocks are low and export demand has been strong through the marketing year. The market is counting on the new crop to ease tightness.
- **December at premium to July (contango)**: old-crop supply is comfortable. Stocks are adequate; no urgency to consume current inventory. This can also occur when the new crop is expected to be small (drought concerns) — the market is paying up for future delivery because new crop availability will be reduced. In this case, the contango is actually a new-crop tightness signal, not an old-crop abundance signal.

- **The December-March spread (within new-crop)**: reflects the pace at which new US cotton moves from farm gate through gin to warehouse and into export. When this spread is in contango (March above December), carry costs and storage economics dominate. When in backwardation (December above March), immediate demand for new-crop is acute.

9.3 Carry and Storage Economics

- **Cost of carrying cotton in certified warehouses**: approximately 25–35¢/bale/month for storage, plus financing (rate-dependent), plus insurance. At 60¢/lb and current interest rates, full carry across a 2-month spread (e.g. December to March) is approximately 150–200 points (0.15–0.20¢/lb). When the spread exceeds full carry, storage arbitrage is available and moderates the spread.
- **Cotton quality degradation**: cotton can degrade in storage if not properly managed (moisture, pest exposure). Certified warehouse stocks maintain quality under controlled conditions, but prolonged storage does add a quality-premium risk that is embedded in the carry cost.

10. Weather, Crop Conditions, and Growing Season Data

Weather is the dominant near-term supply variable for CT futures. Two distinct weather windows matter: the US growing and harvest season (April–November), and the Indian monsoon season (June–September). The US Texas High Plains drought risk is the most acute short-cycle supply variable in the CT market.

10.1 US Crop Progress and Condition Reports

- **USDA NASS Weekly Crop Progress**: published every Monday afternoon during the growing season. Reports: (1) percentage of cotton planted vs year-ago and 5-year average; (2) percentage emerged; (3) crop condition ratings (Excellent/Good/Fair/Poor/Very Poor). The crop condition percentage Good+Excellent is the primary weekly supply quality indicator. A Good+Excellent rating below 35% in late July-August historically correlates with below-average final yields.
- **US Drought Monitor**: published weekly by NOAA/NDMC. The drought severity classification (D0–D4) in Texas High Plains counties (Lubbock, Hale, Swisher, Castro, Floyd) is the most closely watched geographic subset for CT traders. D3 (extreme) and D4 (exceptional) drought coverage above 50% of THP acres in July-August is an acute bullish signal for the nearby CT contract.
- **Abandonment rate projections**: once NASS surveys and county reports indicate widespread stand failure, the market begins estimating abandonment rates — the percentage of planted acres that will not be harvested. Historical abandonment rates: 15–20% in normal years; 40–70%+ in severe drought years (2011, 2022). Each 1 million additional abandoned acres reduces US production by approximately 350,000–450,000 480-lb bales.
- **NOAA COOP station and RAWS weather data**: used to track soil moisture, temperature, and precipitation at individual county level in Texas. Commercial crop scouts also provide field-level condition assessments.

10.2 Bollgard / Bt Technology and Pest Management

- **Bt (Bacillus thuringiensis) transgenic cotton**: provides inherent resistance to lepidopteran bollworms (tobacco budworm, cotton bollworm, pink bollworm). Essentially all US cotton (>98%) is Bt. However, secondary pests (aphids, thrips, whitefly, stink bugs) are not controlled by Bt and require insecticide management. Pest pressure affects yield and quality.
- **Herbicide-resistant weeds**: Palmer amaranth and other weeds resistant to glyphosate have become a significant production challenge in the US Cotton Belt, increasing production costs and potentially reducing planted area in affected counties.

10.3 ENSO and Cotton Production

- **El Nino and the US Southern Plains**: El Nino typically brings above-average winter rainfall to the southern US, which can partially recharge soil moisture in the Texas High Plains ahead of planting — modestly bullish for the new-crop yield outlook.
- **La Nina and US drought**: La Nina is strongly associated with below-average rainfall in the southern US in winter and spring — one of the most consistent US regional climate signals. La Nina entering the US cotton growing season increases drought risk for the THP significantly and is a structural bearish signal for new-crop production.
- **ENSO and Indian cotton**: as for other Indian agricultural commodities, El Nino weakens the SW Monsoon and is bearish for Indian cotton yields. La Nina strengthens the monsoon and is bullish.

11. Macro, Trade Policy, and Cross-Market Drivers

11.1 The USDA WASDE — The Monthly Balance Sheet

The USDA World Agricultural Supply and Demand Estimates (WASDE), published monthly on approximately the 11th of each month, is the single most important scheduled report for CT futures. The cotton section provides:

- US and world production, mill use, trade, and ending stocks — for both the current and prior marketing years. The US ending stocks-to-use ratio (end-of-year stocks as a percentage of annual mill use + exports) is the primary structural price indicator for CT.
- **World stocks-to-use ratio**: the ICAC (International Cotton Advisory Committee) tracks global stocks-to-use. When world S/U falls below approximately 70–75%, CT tends to trade above its long-run average. Above 80–85%, CT tends toward the lower end of the historical range. Note: China's state reserve cotton is included in world stocks but is not freely available to the market — many analysts track "world stocks ex-China" as a more accurate tightness measure.
- **August WASDE**: the first official US cotton production estimate for the new crop (based on August 1 conditions survey). This is the most market-moving WASDE of the crop year for CT, because it converts weather and planting data into the first quantified supply number.

11.2 US-China Trade Policy

- **Section 301 tariffs and retaliatory tariffs**: the US-China trade dispute initiated in 2018 included 25% retaliatory tariffs by China on US cotton imports. These tariffs effectively closed the Chinese market to US cotton for an extended period, redirecting US exports to Bangladesh and Vietnam and redirecting Chinese imports to Brazil and Australia. This trade flow disruption affected: (1) the pace of US export sales and cumulative shipments; (2) the differential between US cotton (CT basis) and non-US origins; (3) the Brazilian and Australian currency-adjusted competitiveness vs US cotton.
- **UFLPA and Xinjiang supply chain**: as described in Section 4, Western brand avoidance of Xinjiang cotton has increased demand for certified non-Xinjiang origins — primarily US, Australian, and Brazilian cotton. This is a structural medium-term supportive factor for CT prices relative to a fully fungible world market.

11.3 The USD and Emerging Market Currencies

- **USD Index (DXY)**: cotton is priced globally in USD. A stronger USD makes US cotton more expensive for foreign buyers, reducing export demand and weighing on CT. A weaker USD makes US cotton cheaper internationally, supporting export sales.
- **Indian Rupee (INR) and Brazilian Real (BRL)**: depreciation of INR and BRL relative to USD makes Indian and Brazilian cotton cheaper in dollar terms, increasing their competitiveness vs US cotton in international markets — a headwind for US export sales and CT prices. This is analogous to the BRL effect on No. 11 sugar.
- **Pakistani Rupee (PKR)**: Pakistan's energy crisis and PKR weakness (acute in 2022–2023) have simultaneously reduced Pakistani mill utilisation (reducing demand for imported cotton) and created economic stress that affects the competitiveness of Pakistan as a textile exporter.

11.4 Crude Oil and Polyester Linkage

- As described in Section 6, the cotton-polyester price ratio is driven by crude oil prices (via PTA/para-xylene economics). When crude oil rises sharply, polyester becomes more expensive, improving cotton's competitive position and supporting CT demand. When crude falls, cheap polyester erodes cotton demand. This creates a partial positive correlation between crude oil prices and CT futures in normal market conditions.

- **Chinese PTA overcapacity**: China has massively expanded domestic PTA production capacity, creating a structural surplus that depresses PTA margins and keeps polyester prices lower than crude oil alone would imply. This persistent overcapacity is a structural bearish factor for cotton demand at the margin.

11.5 Speculative Positioning — COT Data

- **CFTC COT for Cotton No. 2**: weekly managed money net position. Cotton is heavily speculated; managed money net positions have historically ranged from approximately −30,000 contracts (extreme net short) to +80,000+ contracts (extreme net long) in response to supply disruptions and demand cycles. Extreme net long positions correlate with elevated reversal risk, particularly when a bullish weather story resolves or a WASDE print is below bullish consensus.
- **Index fund rebalancing**: cotton is included in commodity indices (GSCI, Bloomberg Commodity Index). Index roll periods (particularly January) create mechanical buying/selling pressure in front contracts.

12. Documented Seasonal Patterns

Cotton has well-documented seasonal patterns driven by the US crop calendar and the marketing year structure. However, these patterns are overridden by weather and policy events more frequently than for some other commodities.

Period	Crop Stage	Typical CT Behaviour	Key Variables	Reliability
Jan–Feb (post-harvest)	Old-crop marketing. Gin runs continuing. Export loading.	Tends to firm as export pace vs USDA forecast is evaluated. Low stocks years: CT can maintain backwardation. Adequate stocks: mild contango, limited volatility.	Weekly export sales and shipments vs USDA forecast; certified stock levels; early planting intentions signals from Texas	Moderate
Mar–Apr (planting intentions)	USDA Prospective Plantings (late March). New-crop planting begins.	Prospective Plantings acreage vs market expectations drives a significant one-day CT move. Below-expected acreage is bullish new-crop; above is bearish.	USDA Prospective Plantings (late March); early planting conditions; soil moisture in THP	High for Prospective Plantings day specifically
May–Jun (emergence)	Cotton emerging and establishing. Squaring approaching.	Crop condition Good+Excellent ratings set the baseline yield outlook. Texas moisture conditions most watched.	USDA weekly Crop Progress; TX drought monitor; ENSO forecast for La Nina risk	Moderate
Jul–Aug (critical boll development)	Boll set; peak heat and drought sensitivity. Texas HP most at risk.	Most volatile period. Weather events dominate. Drought/abandonment fears move CT 3–8¢ in days. First WASDE production estimate (August) is year's biggest single scheduled event.	USDA Crop Progress weekly; US Drought Monitor; August WASDE; NASS August crop report	High — weather dominated, but directionally obvious when drought is severe
Sep–Oct (harvest begins)	US harvest commencing in THP and Southeast.	As harvest data confirms production estimates, CT tends to find equilibrium. Old-crop/new-crop spread (July-December) narrows or reverses.	USDA weekly harvest progress; NASS ginning data; US weekly export sales (new-crop bookings)	Moderate
Nov–Dec (full harvest, new-crop marketing begins)	US harvest concluding. Gin runs at peak. Export loadings beginning.	December CT approaches expiry: certified stock vs open interest determines squeeze risk or delivery abundance. End-of-year positioning adjustments add mechanical volatility.	Certified stock vs December open interest; weekly export sales; Cotlook A basis	Moderate — December expiry mechanics add volatility near expiry

12.1 Key Documented Historical Events

- **2010–2011 record rally**: CT peaked above 227¢/lb in March 2011 — the highest price in the contract's history. Driven by: simultaneous production shortfalls in the US, Pakistan (severe flooding 2010), and China; strong Chinese import demand; strong apparel demand recovery post-GFC; and excessive speculative positioning. Followed by one of the sharpest reversals in soft commodity history.
- **2011 Texas drought**: US production fell from approximately 18 Mb to 15.6 Mb. Texas abandonment exceeded 70% in some counties. CT peaked above 200¢/lb before the drought was confirmed.
- **2018 US-China trade war**: China's 25% retaliatory tariff on US cotton effectively removed China from the US export market. US export pace fell sharply; the Cotlook A-CT basis widened as US cotton was displaced. CT declined substantially as US ending stocks built.
- **2022 Texas drought and Pakistan floods**: simultaneous production disruption in the US (drought) and Pakistan (devastating monsoon floods). CT reached 153¢/lb before demand destruction (polyester substitution, weak apparel orders) capped the rally.

13. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Release Timing	Primary CT Curve Impact
USDA WASDE — Cotton component	USDA WAOB	Monthly	~11th of month, 12:00 ET	All contracts — August WASDE (first new-crop estimate) is the most critical; US ending stocks-to-use ratio is the structural anchor
USDA NASS Prospective Plantings	USDA NASS	Annual	Late March (last business day)	December (new-crop) — acreage vs expectations sets the supply trajectory for the marketing year; among the largest single-day CT moves
USDA NASS June Acreage Report	USDA NASS	Annual	Late June	December — confirms or revises planted acreage from Prospective Plantings; updates abandonment expectations
USDA NASS Weekly Crop Progress	USDA NASS	Weekly (Apr–Nov)	Monday ~4:00 PM ET	Front 2 contracts — Good+Excellent rating in Jul-Aug most market-moving; planting/harvest progress pace
USDA NASS August Crop Report	USDA NASS	Annual	Released with August WASDE	December — first yield estimate by state; Texas HP yield is the primary number
USDA Weekly Export Sales Report (ESR)	USDA FAS	Weekly	Thursday 8:30 AM ET	C1–C3 — upland cotton net sales and shipments by destination; China and Bangladesh most watched
US Drought Monitor	NOAA / NDMC	Weekly	Thursday morning	Front contracts (Jul-Aug) — Texas High Plains drought severity; D3/D4 coverage is the acute bullish trigger
USDA NASS Ginning Reports	USDA NASS	Monthly (Oct–Mar)	~3rd week of month	C1–C2 — actual ginned bales confirm production vs WASDE estimate
ICE Certified Cotton Stocks	ICE Futures US	Daily	After close	Front contract — low certified stocks approaching expiry = delivery squeeze risk
Cotlook A and B Indices	Cotlook Ltd	Daily	Business hours UK	C1–C3 — global physical cotton benchmark; Cotlook A-CT basis indicates US export competitiveness
ICAC World Cotton Outlook	International Cotton Advisory Committee	Monthly	~2nd week of month	C3–C6 — global production, mill use, trade, stocks; world S/U ratio (ex-China most relevant)
NOAA ENSO forecast	NOAA CPC	Monthly	~Mid-month	C4–C8 — La Nina risk for Texas HP drought; El Nino signal for Indian monsoon
IMD India Monsoon Forecast	India Meteorological Dept.	Seasonal + monthly	April first; monthly updates	C4–C6 — Indian production outlook; affects WASDE India estimates 4-6 months later
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET	Positioning / crowding — all contracts; extreme managed money positioning as contrary indicator
BGMEA Bangladesh Garment Export Data	BGMEA	Monthly	~2nd week of month	C2–C5 — garment export orders are the leading indicator of Bangladeshi cotton import demand

ABARES Australian crop estimates	ABARES / Cotton Australia	Seasonal	Quarterly	C3–C5 — Australian counter-seasonal supply; relevant for Northern Hemisphere off-season contracts
CCI India procurement data	Cotton Corp. of India	Weekly (season)	During Indian marketing season	C2–C4 — MSP procurement pace indicates whether Indian market prices are at or below MSP; signals export availability

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ICE Cotton No. 2 is a market defined by three structural dualities: natural fibre vs synthetic fibre competition (cotton vs polyester); US price-setting supply vs rest-of-world production dynamics; and the Chinese state apparatus simultaneously as consumer, importer, producer, and reserve manager. No other soft commodity has the same complexity of cross-commodity price linkages (to crude oil via polyester), trade policy dependencies (US-China tariffs, Xinjiang supply chain), and weather concentration risk (Texas High Plains dryland production).

For spread traders, the most reliable edges come from three recurring patterns: (1) the Prospective Plantings acreage surprise in late March — when US farmers intend to plant significantly less cotton than expected, the December contract reprices immediately and the move is typically sustained; (2) the Texas High Plains drought signal in July-August — when the Drought Monitor shows D3/D4 coverage above 50% of THP acres, the abandonment rate will be high and the US crop small, a rare but extreme bullish signal that moves CT faster and further than almost any other agricultural commodity weather event; (3) the old-crop/new-crop spread (July-December) — a reliable structural expression of whether current marketing-year stocks are adequate or tight relative to the incoming new crop, incorporating the full supply and demand picture at each end of the old/new crop divide.